

Result: full rollouts retain better precision

Current default: exp277, step 266344. Eval-val: 97 natural proteins. 100 full rollouts:
R-precision 0.55743 all / 0.54149 long. Best short arm: 1,000 rollouts capped at
floor(L/2): 0.55003 / 0.53534. All-range delta: -0.00740; paired 95% CI [-0.01094,
-0.00396]. Every tested short cap loses precision. No eval-test scoring.

What was compared

Five caps: 5, 10, 20, $\text{floor}(L/5)$, $\text{floor}(L/2)$ emitted contact statements. 1,000 independently resampled prompts; smaller caps reuse trajectory prefixes. 100-short subsets control for sample count. A fresh 100-full baseline. Temperature 1, top-p 0.95, top-k off. Same candidate universe and exp89 metrics. 97,000 short and 9,700 full samples; zero unfinished full rollouts. All 1,067 vote matrices reconstructed from raw samples and verified.

AUC improves, at higher cost for the largest cap

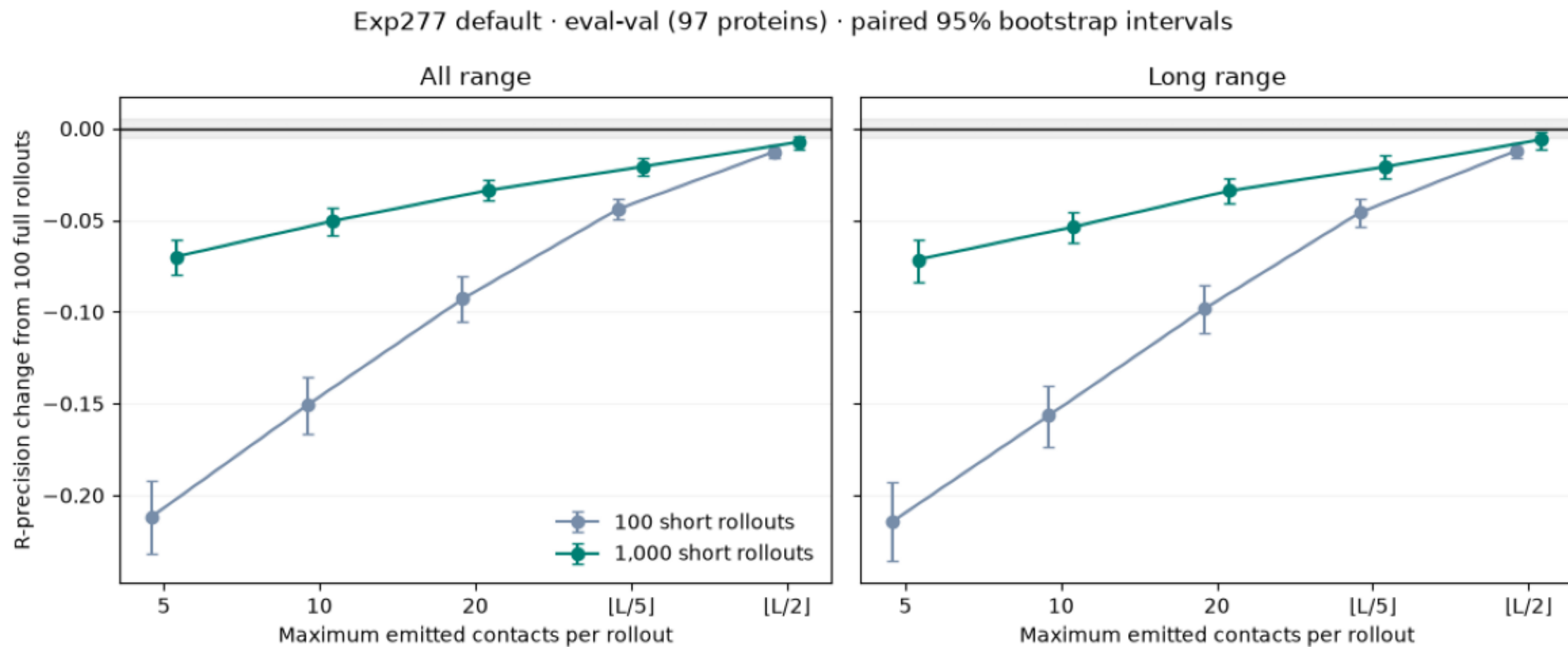
At L/2: all-range AUC 0.93921 -> 0.95565; long AUC 0.92872 -> 0.94849. Global ranking improves while highest-ranked-contact precision worsens. Shared short generation: 25.0 minutes; full baseline: 6.9 minutes (3.60x). L/2 uses 4.98x the output tokens and 10x the prompt tokens. Smaller-cap standalone runtimes were not measured; their prefixes share generation.

Scope and recommendation

Keep 100 full rollouts for contact precision under this tested recipe. The fresh baseline is within 0.005 of the published exp277 baseline. KNN reference: 0.40715 all / 0.39211 long; all 1,000-short arms clear it. Six viral proteins: same direction, small subgroup. Low-MSA cut: zero coverage. One sampling seed on eval-val; no held-out generalization claim. Raw samples, vote matrices, tables, timings, and runtime provenance are public.

paired_rprecision.png

All five caps, with 100-short controls; shading marks ± 0.005 . Intervals are exploratory and pointwise.



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$ analyze.py --results ../../scratch/exp354/production
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